CS 180 • PROJECT 4 • FALL 2026

[Auto]Stitching Photo Mosaics

From hand-checked point pairs to automatically recovered panoramas: a reproducible study of homographies, inverse warping, feature descriptors, robust matching, and cylindrical projection.

Part A • 17 annotated pairs **Part B • Harris + ANMS + RANSAC**

20 source frames • all used **Three complete panoramas**

INTRODUCTION

Rotate, register, blend

The same scene appears at different image coordinates as a camera turns. Part A estimates this relationship from 16 manually verified landmarks in every adjacent pair; Part B finds those landmarks automatically. Every panorama shown below uses all frames of its sequence. Compact planar spans are retained as a controlled homography baseline, while a cylindrical output accommodates each full, wider sweep.

CORE GEOMETRY

One point, three coordinates

A perspective warp maps a homogeneous source point p to a destination point p' using a 3×3 homography. Inverse mapping assigns each output pixel a source sample and avoids holes.

$$p' \sim H\,p \quad \cdot \quad p = (x, y, 1)^T$$
$$I_{out}(q) = I_{src}(H^{-1}q)$$

EXPERIMENT MAP

Two ways to find the geometry

Part A	Capture three overlapping sequences; hand-check correspondences; solve least-squares homographies; implement two samplers, two rectifications, and feathered mosaics.
Part B	Detect Harris corners, apply ANMS, form normalized 8×8 descriptors, ratio-match, and fit four-point RANSAC homographies before mosaicing.
Extension	Inverse-project all frames onto a cylinder when the field of view is too wide for one stable planar canvas.

PART A

Image warping and mosaicing

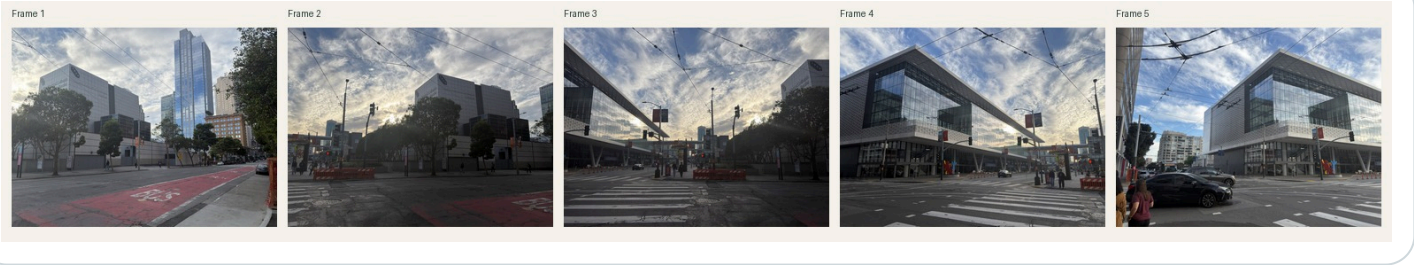
All point pairs, homographies, rectifications, and images below are generated from the included resized inputs. The original full-resolution captures remain in the project workspace.

A.1 • SHOOT AND DIGITIZE PICTURES

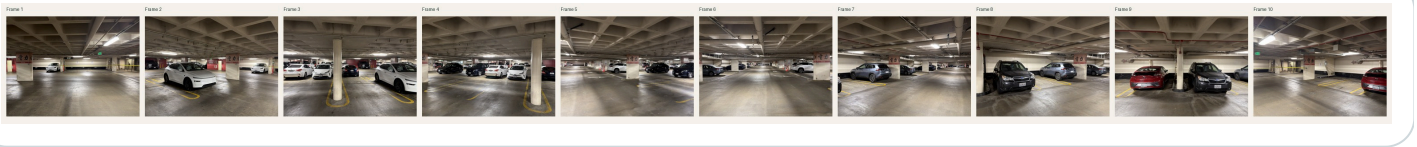
Three overlapping capture sequences

5 + 10 + 5 frames

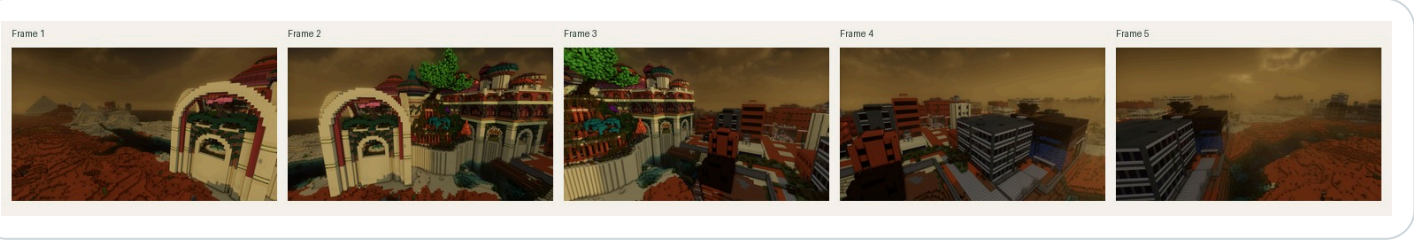
The street and garage are the two primary real-camera sets. The virtual set supplies a third architectural case with controlled camera rotation and strong shared structures. Neighboring views overlap around buildings, columns, and the decorated gate; sky and moving objects are not used as primary landmarks. Every frame below participates in a full panorama later in the report.



Outdoor street • frames 1–5. One turning sequence around a city intersection and large glass building.



Indoor garage • frames 1–10. A much wider turn with repeated columns, cars, and ceiling beams.



Virtual architecture • frames 1–5. Shared building details, rather than changing clouds, anchor the correspondences.

A.2 • RECOVER HOMOGRAPHIES

Sixteen constraints per adjacent pair

17 pairs checked

The final JSON contains 16 landmark pairs for each of the 4 outdoor, 9 garage, and 4 virtual transitions. The manually dragged virtual edits are preserved; one imprecise destination point was snapped 10 px left and 4 px up after comparing its local texture. This is a local correction, not a replacement of the user's large moves.



Outdoor. Corners of facades and street structure.



Indoor. Cars, columns, and ceiling intersections.



Virtual. The same gate and facade details across the turn.

LINEAR SYSTEM

Overdetermined least squares

Fix $h_{33}=1$. Each correspondence $(x,y)\to(u,v)$ contributes two rows, so 16 pairs give a 32×8 system. I normalize each point set, solve the eight unknowns with least squares, then denormalize H.

```
[x y 1 0 0 0 -ux -uy] h = u
[0 0 0 x y 1 -vx -vy] h = v
               A h = b
```

MEASURED EXAMPLE

Outdoor frame 1 → 2

```
H = [[ 0.424727,  0.048149, 422.304858],
     [-0.258685,  0.781745, 108.137303],
     [-0.000499, -0.000046,   1.000000]]
```

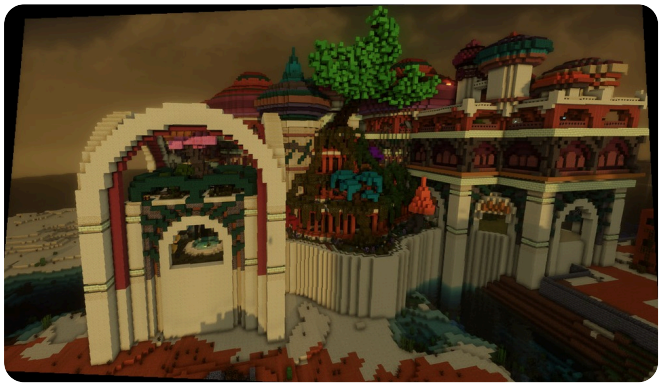
Sixteen points give a 2.35 px RMS reprojection error. Across all 17 manually annotated pairs, the RMS range is 0.49–2.80 px. The measurement file contains every fitted matrix and error.

A.3 • WARP THE IMAGES

Inverse sampling and two rectifications

NN + bilinear

For each output pixel, the inverse homography identifies a floating-point source coordinate. My nearest-neighbor routine rounds to one pixel; my bilinear routine weights the surrounding four. Both are implemented directly with NumPy indexing. On one 1080×624 local run, nearest sampling took 0.153 s and bilinear 0.267 s. Bilinear reduces staircase edges, while nearest is cheaper and preserves block boundaries.



Nearest neighbor. One sample per destination pixel.



Bilinear. Weighted combination of four samples.



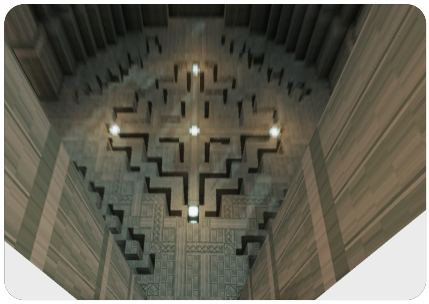
Identical region enlarged to make the interpolation difference visible.

Rectification 1 · toward the reference viewpoint

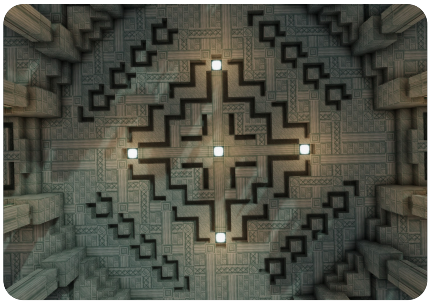
The original 2240×1294 oblique image contains five floor lights also visible in the independent overhead reference. I manually paired the three center lights and the two side lights, fitted an overdetermined floor homography, and warped the original pixels into the reference coordinate system. The panels below use the same reference crop. The center cross and lights now have the reference's orientation and approximate positions; no reference pixels are copied into the reconstruction.



Source landmarks. Three center lights and two side lights determine the ground-plane transform.



Reference-aligned reconstruction. Original source pixels only; pale corners are outside the camera's visible coverage.

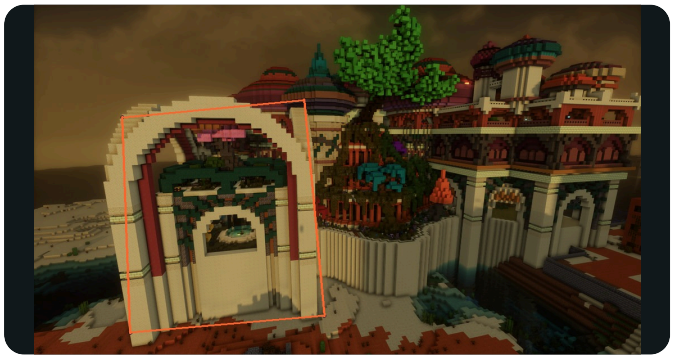


Matching reference crop. A different camera view, shown to compare the central cross and light positions.

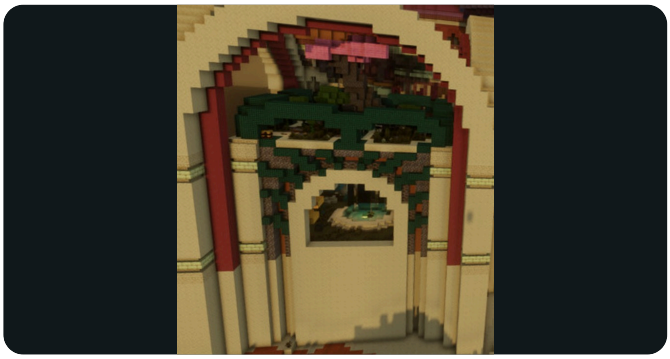
Geometric limit. A homography can correct the floor plane, but the upright columns and raised blocks sit at other depths and stretch under that same transform. The overhead view reveals surfaces hidden in the oblique photo, so it cannot be reconstructed exactly from this single image. The earlier floor trapezoid and its rectified result remain as a conventional four-corner baseline.

Rectification 2 · the complete angled gate

The first attempt used a partly cropped gate and did not straighten it. Here the fully visible second virtual frame supplies all four facade corners. Mapping those corners to a rectangle makes the side posts vertical and the upper frame level. The curved arch itself remains curved, as its shape belongs to the building.



Source, virtual frame 2. The selected quadrilateral encloses the full gate.



Rectified gate. A 650×720 front-facing facade with nearly vertical columns.

A.4 • BLEND INTO MOSAICS

Three complete manual panoramas

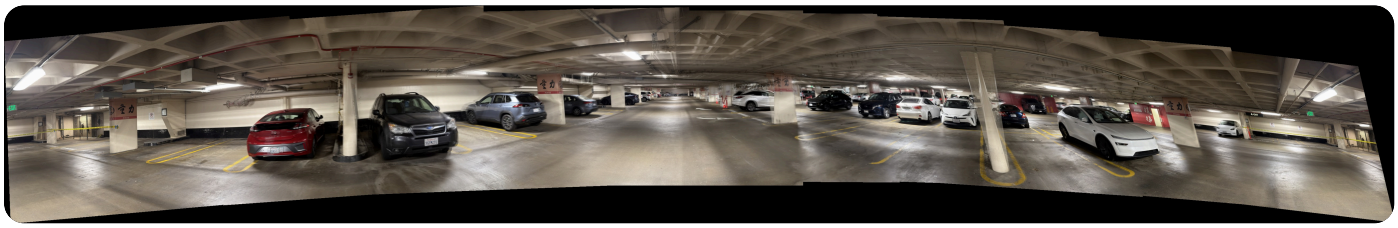
5 • 10 • 5 frames

Pairwise transforms are composed into a middle anchor frame. Each image is sampled once on the shared canvas. Overlap colors use a source-edge feather weight, then are divided by the sum of weights. Black borders indicate uncovered rays. The input strips in A.1 show every source frame for the three mosaics below.

$$C(q) = \Sigma_i \alpha_i(q) I_i(q) / \Sigma_i \alpha_i(q)$$



Outdoor • all 5 frames. Center frame 3 is the anchor; the complete street sweep extends from the close building to the opposite curb. Click for full resolution.



Indoor • all 10 frames. The ten-frame garage sweep covers the left and right parking rows and the long central aisle.



Virtual • all 5 frames. Architecture connects the surrounding landscape and town buildings.

What the overlap reveals. The center of each scene is coherent; residual ghosts appear at nearby cars, columns, and objects with depth. Feathering softens boundaries but does not undo parallax.

A.5 • BELLS AND WHISTLES

A cylinder makes wide turns finite

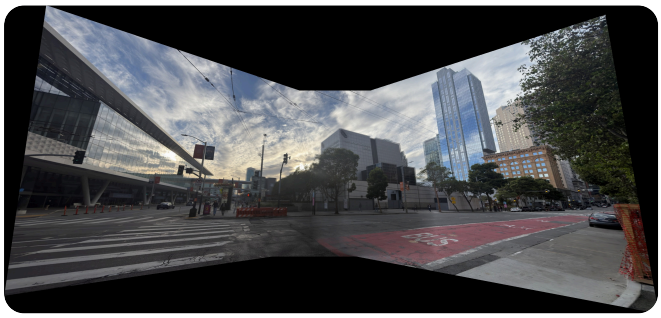
inverse projection

A flat homography canvas becomes unstable at the edges of a wide rotation. The full mosaics in A.4 therefore use a cylindrical target. For each output cylinder pixel, I invert the cylindrical ray to sample the original image directly, then fit a small affine correction to the projected manual points. Focal lengths were selected by minimizing adjacent affine reprojection error on a small grid and checking the visual shape.

$$x_{cyl} = f \operatorname{atan}((x-c_x)/f) \quad \cdot \quad y_{cyl} = f(y-c_y)/\sqrt{((x-c_x)^2+f^2)}$$

Outdoor f = 775 px | Indoor f = 825 px | Virtual f = 450 px | Valid center rectangle only

The original projective homography pipeline also produces these compact three-frame planar baselines. They show why a flat target is useful over a moderate range; the full cylindrical results above use every frame.



Outdoor planar baseline • frames 1–3. Direct H composition and feathering.



Garage planar baseline • frames 4–6. A short span avoids extreme projective stretching.

PART B

Automatic feature matching

No manual point coordinates enter the Part B path. The same images are processed by Harris detection, ANMS, MOPS-style descriptors, ratio matching, and four-point RANSAC before each full sequence is stitched.

B.1 • HARRIS AND ANMS

Find corners, then spread them out

single scale

I blur grayscale intensity, compute x/y derivatives, and score the second-moment matrix with $R = \det(M) - 0.04\ \mathrm{tr}(M)^2$. Harris local maxima cluster around textured regions. Adaptive nonmaximum suppression ranks each corner by distance to a significantly stronger neighbor, increasing spatial coverage before matching.



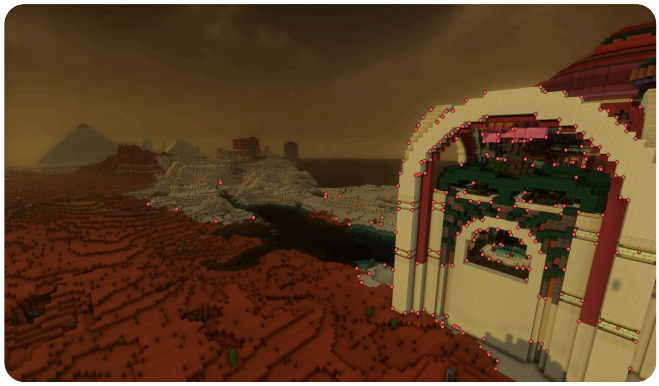
Before ANMS. 450 detected outdoor corners, concentrated in high-contrast structure.



After ANMS. A 220-point visualization budget makes the spatial redistribution clear. Actual matching uses up to 550 selected points.



Garage: selected corners span ceiling, cars, and columns.



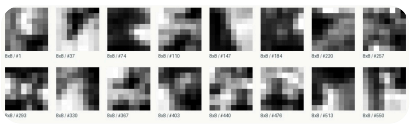
Virtual scene: corners favor the gate and building edges.

B.2 • FEATURE DESCRIPTORS

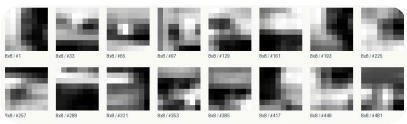
Normalize local appearance

40×40 → 8×8

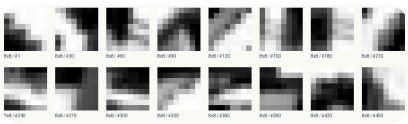
A blurred 40×40 axis-aligned support is reduced to an 8×8 descriptor. Subtracting its mean removes additive brightness; dividing by its L2 norm reduces contrast dependence. This follows the simplified patch descriptor requested by the course rather than a rotation-invariant version.



Outdoor. Enlarged 8×8 patch examples.



Indoor. Patch diversity despite repeated beams.



Virtual. Block edges and facade texture.

B.3 • FEATURE MATCHING

Appearance suggests candidate pairs

ratio = 0.78

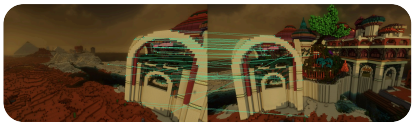
For each descriptor I compare its first and second nearest neighbors, retain it when the distance ratio is below 0.78, and require a mutual nearest match. The result is still imperfect; geometry provides the final check. The representative panels below show pre-RANSAC candidates from each sequence.



Outdoor pair 1: 41 tentative matches.



Garage pair 4: 115 tentative matches.



Virtual pair 1: 46 tentative matches.

B.4 • FOUR-POINT RANSAC

Reject outliers; stitch all frames

3 automatic mosaics

Each trial samples four matches, estimates H with my least-squares solver, and counts points whose forward reprojection error is below 4 px. The best set is refitted on all inliers. The same inliers are projected onto the cylinder for full-sequence placement. No high-level homography, perspective warp, or RANSAC function is called.

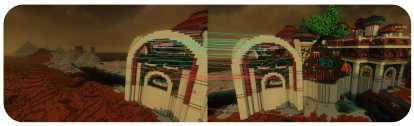
4 sampled matches | 4 px inlier radius | Up to 2400 trials | Refit all inliers | Feathered final blend



Outdoor. 34 of 41 representative matches remain.



Indoor. 107 of 115 representative matches remain.



Virtual. 27 of 46 representative matches remain. Green denotes an inlier; coral denotes an outlier.

All adjacent pairs were processed

OUTDOOR • 5 FRAMES

Pair	Matches	Inliers
1→2	41	34
2→3	157	130
3→4	122	89
4→5	97	72

GARAGE • 10 FRAMES

Pair	Matches	Inliers
1→2	61	53
2→3	91	68
3→4	125	98
4→5	115	107
5→6	64	47
6→7	194	152
7→8	181	157
8→9	97	64
9→10	42	19

VIRTUAL • 5 FRAMES

Pair	Matches	Inliers
1→2	46	27
2→3	58	44
3→4	15	9
4→5	37	21

The narrow 3→4 overlap is hardest, yet yields nine geometric inliers.

Manual and automatic full-sequence results

The following pairs use identical source images and cylindrical projection. Only the correspondence source changes. Click either result for the full-resolution image; this matters especially for the ten-frame garage panorama.

Outdoor · 5 of 5 frames

MANUAL



Four transitions, 16 checked landmarks each; 775 px cylinder.

AUTOMATIC



Four automatic RANSAC fits; the overall street geometry follows the manual result.

Indoor garage · 10 of 10 frames

MANUAL



Nine manually fitted transitions; 825 px cylinder.

AUTOMATIC



All nine transitions pass RANSAC; the final pair retains 19 inliers.

Virtual architecture · 5 of 5 frames

MANUAL



Full architectural sweep; 450 px cylinder.

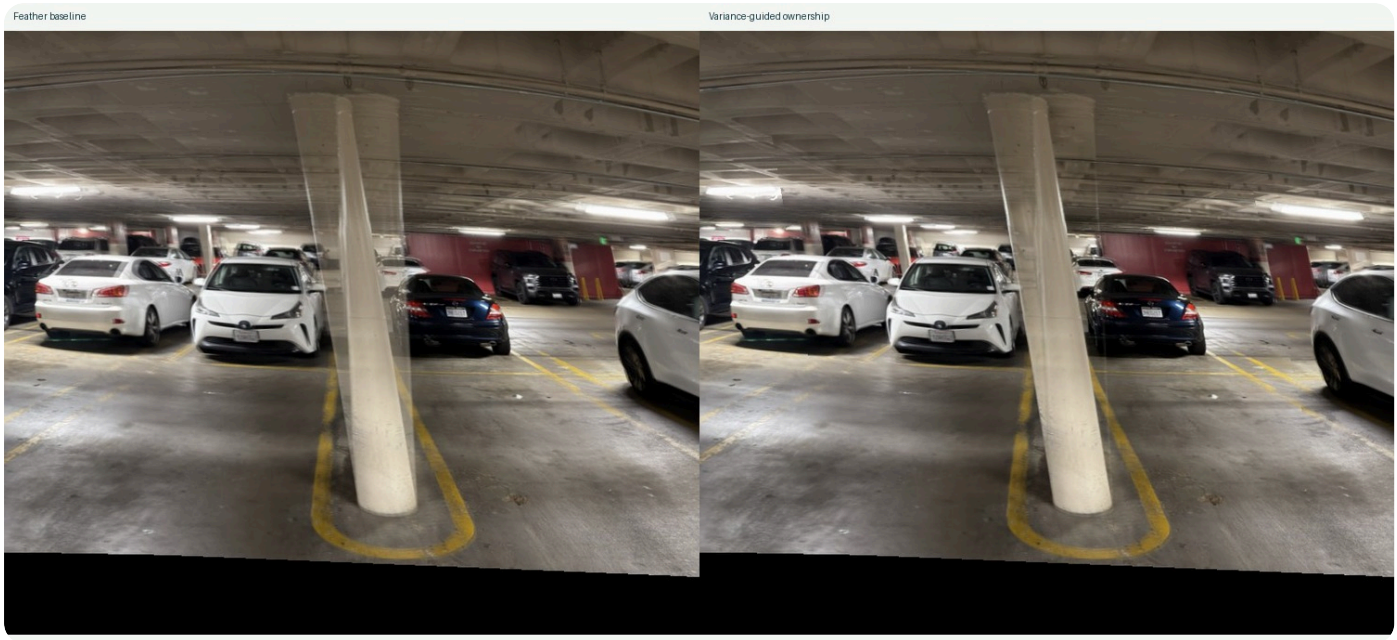
AUTOMATIC



Architecture remains continuous in both versions; the far terrain has less geometric detail.

Why the garage column ghosts

The near white column shifts against the distant parked cars between overlapping camera views. RANSAC correctly aligns the dominant background, but one global transform cannot also align the column at a different depth. Feather blending averages both projected column edges, producing a translucent double contour. This is mainly capture parallax exposed by the global-warp and blending model, rather than a failed correspondence fit.



Diagnostic. Left: required feather blend. Right: a variance-guided dominant-source blend sharpens parts of the car and pillar, but a second pillar edge remains. A local seam, local warp, or depth-aware alignment would be needed to remove it reliably; we retain feathering as the required baseline.
Open the full diagnostic mosaic.

Comparison. The main structures agree between the manual and automatic panoramas. Differences are concentrated near depth changes, moving cars, and repetitive texture. A single global warp per frame cannot correct parallax, and simple feathering cannot remove every ghost. The full-view cylinder avoids the severe edge stretching that limited the earlier three-frame-only report.

CONCLUSION

What the panorama needs

A + B complete

Reliable correspondences

Sixteen checked points per pair give a manual reprojection RMS range of 0.49–2.80 px. Automatic detection and descriptor matching recover enough geometrically consistent points in all 17 pairs.

The right target surface

Planar homographies work for moderate spans and rectification. A cylinder keeps all 20 input frames useful across three wide sweeps without forcing a single unstable planar canvas.

Visible limits

The deeper floor rectification enlarges distant pixels and loses sharpness. Nearby objects create parallax ghosts. These are source-resolution and scene-geometry limits, not missing interpolation samples.

REPRODUCIBILITY

Run from the archive root

```
python code/main.py --data-root code/data --
output results
```

The archive contains the resized source frames, final correspondence JSON, Python implementation, this offline HTML report, and a printed PDF. The metrics JSON records all fitted matrices, per-pair counts, and mosaic canvas sizes.

SOURCES

Course and method

Official Part A · Official Part B · Brown, Szeliski, and Winder: Multi-Image Matching Using Multi-Scale Oriented Patches.

The descriptor implementation uses the axis-aligned 8×8 simplification specified for this project.